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Beyond empathy: Cognitive capabilities increase or curb altruism in middle childhood



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ABSTRACT

Altruistic behavior, which intentionally benefits a recipient without expectation of a reward or at a cost to the actor, is observed throughout the lifespan from everyday interactions to emergency situations. Empathy has long been considered a major driver of altruistic action, but the social information processing model supports the idea that other cognitive processes may also play a role in altruistic intention and behavior. Our aim was to investigate how visual analysis, attention, inhibitory control, and theory of mind capabilities uniquely contribute to predicting altruistic intention and behavior in a sample of 67 French children (35 girls and 32 boys; $M_{\text{age}} = 9.92 \pm 0.99$ years) from Paris and neighboring suburbs. Using a Bayesian analysis framework, we showed that in younger grade levels visual analysis and selective attention are strong predictors of altruistic intention and that inhibitory control strongly predicts altruistic behavior in a dictator game. Processes underlying theory of mind, however, negatively predict altruistic behavior in the youngest grade. In higher grade levels, we found that stronger attention and inhibitory control predicts lower altruistic intention and behavior. Empathy was not found to predict altruistic intention or behavior. These results suggest that different cognitive capabilities are involved in altruistic intention and behavior and that their contribution changes throughout middle childhood as

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social constraints deepen and altruism calls on more complex reasoning.

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Introduction

Educators and parents regularly encourage children to find practical applications to Charles Dickens' assertion that "no one is useless in this world who lightens the burdens of another" (Dickens, 1894) by comforting a sibling or explaining a math problem to a classmate. Yet, lightening the burden of another with nothing in return, or even at a cost to oneself, is another situation entirely. Altruistic behavior is considered as voluntary and intentional behavior that aims solely at benefitting another without expectation of an external reward or to avoid external punishment (Bar-Tal, 1985; Eisenberg & Miller, 1987). It is a more constraining subset of prosocial behavior, which is behavior that voluntarily and intentionally benefits others but can also include actions with reciprocation or rewards. Prosocial behavior and altruistic behavior can be expressed in various ways, including helping, sharing, and comforting others in response to perceived instrumental needs to achieve a goal, unmet material desire, and emotional distress of another, respectively (Dunfield, 2014; Dunfield & Kuhlmeier, 2013; Rose et al., 2023). The influence of social context and social-moral norm activation on the implementation of altruistic actions has been studied extensively, notably in behavioral economics with a classic game theory paradigm known as the dictator game in which a participant receives a windfall endowment and can unilaterally decide how much to share with a recipient, with the latter having no say in the transaction (Blake & McAuliffe, 2011; List, 2007). The underlying cognitive and social-cognitive drivers of this behavior, however, have received comparatively little attention except for empathy as presented below.

The empathy-altruism hypothesis in the spotlight

Empathy includes an affective response to another's emotional experience, also defined as affective "resonance" (Eisenberg et al., 2010), as well as the understanding of a person's feelings without having those feelings resonate personally (Jones et al., 2010). Batson's empathy-altruism hypothesis underlines that empathy drives altruistic action more strongly than personal distress or other negative states (Batson et al., 1988; Cialdini et al., 1987; Estrada, 1995; Hoffman, 2000). Research with children has found positive relations between empathy induction or scale-based measures of empathy, on the one hand, and various forms of altruistic behavior, including donations and prosocial lying, on the other (Malti et al., 2009; Nagar et al., 2020). However, more recent and nuanced results underline methodological challenges but also point to empathy as a driver of altruistic behavior mainly toward those we know (Bloom, 2017; McCauley et al., 2022; Schaefer et al., 2021). These results highlight that research must look beyond the empathy-altruism hypothesis, and toward cognition in general, to get a fuller picture of how altruistic intentions and behavior are enacted. Focusing on childhood, where prosocial milestones echo cognitive development, provides evidence of the many cognitive abilities involved in altruism.

Parallel developments in cognition and altruism

Altruistic behavior has been consistently observed early in development; past their first birthday and around 18 months of age, infants display helping, sharing, and comforting behaviors regardless of external praise or reward (Hepach et al., 2012; Song et al., 2020; Warneken & Tomasello, 2009). Various studies have pointed out the role of different cognitive abilities in triggering altruistic behavior from infancy, where intention understanding positively predicts later instrumental helping in the second year of life, and initiating joint attention in infancy also positively predicts later sharing (Stout

et al., 2021). As children reach 3 years of age and become attuned to subtle cues as well as another's emotional distress, their prosocial behavioral palette becomes more complex (Dunfield et al., 2011; Paulus, 2018; Svetlova et al., 2010; Wellman & Liu, 2004). Norm-based considerations and moral judgments then add to this palette, with children by 10 years of age actively giving more in dictator and ultimatum games (where an endowment is split between two players and the recipient can choose to refuse the split, bringing everyone's total to zero) and also rating happier emotions when doing so (Benenson et al., 2007; Fehr et al., 2008; Kogut, 2012; Smith et al., 2013). Concurrently, friendship becomes a driving consideration in altruistic giving, relying on shared values and the ability to juggle among fairness norms, relationships with peers, and reputational concerns when allocating resources (Heyman et al., 2014; Maunder & Monks, 2019; Paulus et al., 2020; Youniss, 1986). This strategic reasoning in social contexts is supported by middle to late childhood with increases in advanced theory of mind (ToM), or recursive reasoning about mental states, and improvements in executive functioning and inhibition in particular (Chasiotis et al., 2006; Devine et al., 2016; Osterhaus & Koerber, 2021; Smogorzewska et al., 2020; Wang et al., 2016). In addition to this review of evidence that altruistic behavior and cognitive abilities develop concurrently, putting the empathy–altruism hypothesis into question also requires looking closely at how other cognitive capabilities drive altruism in development.

The social information processing model and unique contributions of cognitive abilities

The social information processing (SIP) model underlines the different and sequential mental operations, including cognitive operations (Fig. 1), that help a child to enact a behavioral response during a social interaction (Crick & Dodge, 1994; Dodge & Rabiner, 2004). This model has been extensively applied to the study of aggressive behavior in children, although its authors also recommend applying it to prosociality's other extreme, altruism, for greater generalization (Crick & Dodge, 1994). Viewing

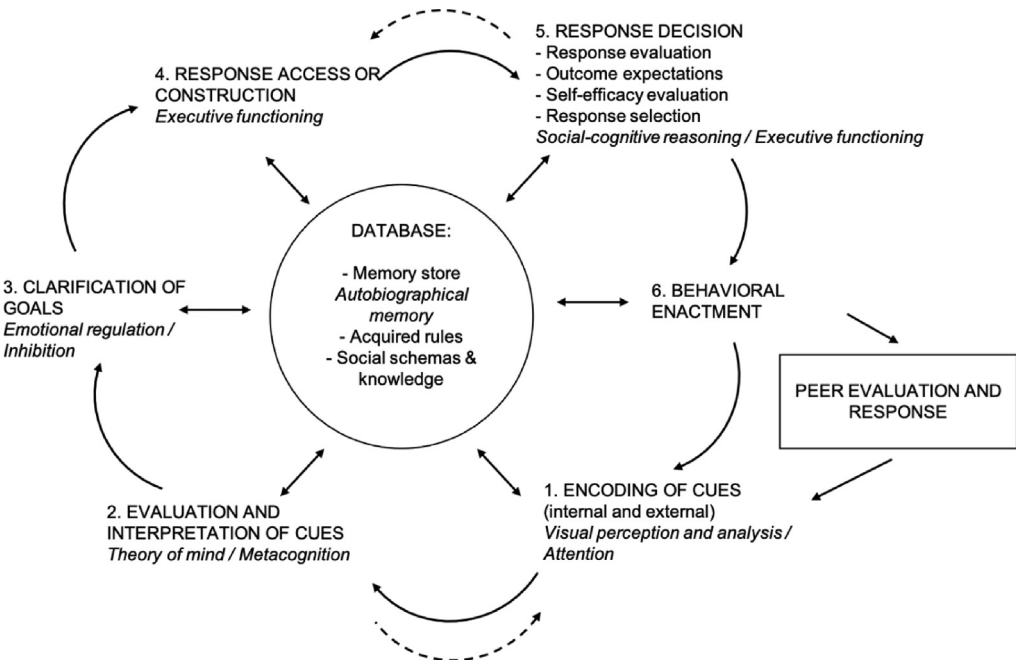


Fig. 1. Crick and Dodge's (1994) social information processing model, including both online processes and latent mental structures and their participation in children's behavior in social situations. Cognitive processes involved at each step were added for illustrative purposes.

the contribution of cognitive processes to altruistic intention and behavior through the lens of the SIP model implies measuring distinct cognitive abilities in a single context (Caporaso et al., 2021), which is the approach we selected for this study.

Most research, however, has focused on the contribution of independent cognitive abilities to altruistic intention or behavior and notably to attentional processes, attentional and behavioral control, and ToM.

Attention, a multifaceted cognitive process, has been linked to prosocial and altruistic behavior throughout childhood. Selective attention to social cues and attention shifting, the ability to select and focus on relevant social stimuli and the ability to shift between them, respectively, both appear to be related to prosocial behavior measured by sharing, and negatively related to aggressive behavior (Schippel et al., 2003; Wilson et al., 2003). From 6 years of age, attentional control, the ability to focus on a task without reduced distraction, is positively related to altruistic giving in a dictator game (Blake et al., 2015). Difficulties in attentional control also predict teacher-reported antisocial behavior in both typically developing children and children with atypical attention abilities from 6 to 12 years of age (Andrade et al., 2009).

Attentional control is closely linked to executive functioning and notably inhibition, the ability to override automatic behaviors and thoughts to implement goal-directed behavior. Studies that looked at inhibition in relation to altruistic behavior have yielded robust results. In younger children aged 4 to 6 years, inhibitory control predicts level of sharing in a dictator game, where children receive a wind-fall endowment and unilaterally decide whether to share it, and these results are also found for older children until 13 years (Aguilar-Pardo et al., 2013; Blake et al., 2015; Steinbeis, 2018; Xie et al., 2019). Interestingly, Hao (2017) found that altruistic sharing relies on different types of inhibition throughout childhood, beginning with 'cool' or decontextualized inhibition as measured by a Stroop task, followed by "hot" or motivation-related inhibition later in middle childhood, measured by a delay of gratification task. As such, altruistic sharing seems to be supported by the ability to inhibit selfish tendencies toward resource hoarding in younger children and by helping children to integrate more complex social and contextual demands in their decisions later.

ToM, which can be defined as a set of cognitive processes or a social competence that allows us to attribute mental states to ourselves and others (Apperly, 2012), also seems to come into play in driving altruistic intent and behavior. Children who perform better on ToM tasks reject unequal offers more often in an ultimatum game, whether the inequality is in their favor or not, and encouraging children to think about others' mental states leads to rejecting offers that are unfair for another but advantageous for themselves, even in younger 6- and 7-year-olds who generally do not display this behavior spontaneously (Tsoi & McAuliffe, 2020). The general trend toward an effect of ToM capabilities on altruistic intent and behavior has been found in various studies that focused on specific facets and measures of the construct, underlining a small but significant correlation ($r = .19$, $p < .05$) (for a meta-analysis, see Imuta et al., 2016).

Despite the importance of various facets of cognition in peer relationships, few attempts have been made to investigate the effect of several cognitive processes at once, most likely due to methodological constraints, and have generally been unsuccessful despite employing various measures of altruism (Liu et al., 2015; Rushton & Weiner, 1975). Most research focuses on one aspect of cognitive development, as detailed above, leading to a limited explanation on a complex construct and the unanswered question of whether certain cognitive capabilities contribute more than others in middle childhood.

The current study aimed to understand the relative contribution of various cognitive abilities, from more perceptive abilities (visual analysis), attentional processes (selective attention and attentional flexibility), inhibition, and social cognitive processes (higher-order ToM), to altruistic intention and behavior in middle childhood. Given past research on the positive impact of empathy on altruistic behavior, we also aimed to compare the contributions of these cognitive abilities and of empathy, assessed using a scale-based measure. In line with recent research on the diversity of altruistic behavior, we decided to include two measures of altruism, namely *altruistic sharing* (through a dictator game design) and scale-based hypothetical situations that we refer to as *altruistic intention* (Rose et al., 2023; Swank et al., 2020). We hypothesized that these cognitive abilities, from perceptual-attentional capabilities to social capabilities (ToM as well as empathy), contribute uniquely to altruistic intention and behavior in middle childhood, mediated by age and grade level.

Method

Participants

A total of 67 children attending the final 3 years in the French elementary school system were included in the study from November 2021 to April 2022. Participants included in the study did not present any neurological or neurodevelopmental disorders, as reported by their parents. Participants ranged in age from 7.75 to 11.94 years ($M = 9.92 \pm 0.99$ years), with mean ages per grade presented Table 1. Recruited participants agreed to take part in a larger 10-week project for which the current study was the first stage, thereby limiting the total possible sample size due to demands on children's and parents' schedules in a context of pandemic-related school closures.

The study included 35 female participants and 32 male participants (52.24% and 47.76%, respectively). Families living in Paris and neighboring suburbs were contacted via local social media networks and using former participant contacts from the university's child research laboratory.

In the initial development stages of our research protocol, we reviewed studies published in the past decade involving young children and multiple cognitive tasks. We aligned our target sample size with those of peer-reviewed studies with robust findings (Asaba & Gweon, 2022; Guinote et al., 2015; Steinbeis, 2018).

In addition to this, employing a Bayesian analysis framework (detailed further in the "Data analysis" section) allowed us to plan an initial analysis after the first wave of participant recruitment. If this analysis yielded strong enough evidence for our hypotheses, we would consider that a second wave of participant recruitment was not needed, in line with the literature (Moerbeek, 2021).

Procedure

Children were accompanied by their parents to the on-site testing quarters at Université Paris Cité. Children were then ushered into an office for a one-on-one session where tasks were presented sequentially. Three task blocks were created, with tasks requiring the strongest attentional effort placed at the beginning of a block; blocks were presented in a randomized order, and a break was offered to participants between each block. As such, measures of altruistic intention and behavior were presented in either the first, second, or third part of the 1-h session. The experimental design for this study was approved by the Université Paris Cité research ethics committee. Both the experimental design and analysis plan were also preregistered before data collection and are available in a public repository ([https://osf.io/cmvtg/?view_only = 769793c0b37b41cebd47b941af06c13f](https://osf.io/cmvtg/?view_only=769793c0b37b41cebd47b941af06c13f)). This repository includes all raw data as well as the analysis script and additional figures, as part of a larger year-long program aimed at understanding the drivers of altruistic intention and behavior in children as well as testing an altruism training program.

Measures

Data on cognitive abilities, altruistic intention, and altruistic behavior were collected using behavioral tasks, self-report scales, and parent-report scales, as detailed below.

Table 1
Participant ages per grade.

Grade level	<i>n</i>	Mean age (years)	<i>SD</i>
Third grade (French CE2)	23	9.01	0.58
Fourth grade (French CM1)	24	10.08	0.70
Fifth grade (French CM2)	20	10.77	0.78

Note. CE2, Cours Élémentaire 2; CM1, Cours Moyen 1; CM2, Cours Moyen 2.

Cognitive and social cognition tasks and scales

Visual analysis. An original visual analysis task was created for this study using classroom posters typically used in the 20th century to encourage descriptions and discussions (images available in the study's public repository, with one example shown in Fig. 2) and selected because of the balance between local details as well as more global elements on the left and right sides of each image.

Children were given 1 min to describe the objects and actions on the image. Responses were recorded, and each item or action was counted as a local or global element on the left or right side of the image. The total visual analysis score was then calculated using the following equation:

$$\text{Visual analysis score} = (\text{total items listed}) \times [1 - (|\text{items listed on left} - \text{items listed on right}| \div \text{total details listed}) \times (\text{local details listed}) \times (\text{global details listed})]$$

Selective attention. The Sky Search subtest of the Test of Everyday Attention for Children (TEA-Ch; Manly et al., 2001) was used to assess selective attention skills. Sky Search is a timed test in which children need to find and circle pairs of identical spaceships among a wide set of distractors.

Attentional flexibility. The Opposite Worlds subset of the TEA-Ch was added to the experiment to assess attentional flexibility and control. In the Opposite World, children are instructed to say the number “one” when reading a number two and vice versa. In the Same World, children read numbers as they appear.

Inhibition. In line with findings on the role of “hot” executive functioning in middle childhood, as reported by Hao (2017), inhibition based on emotional stimuli was assessed using the Happy–Sad Task (Lagattuta et al., 2011). In this task, 20 cards, comprising 10 happy face cards and 10 sad face cards, were presented to participants in a random order. Participants were instructed to say “happy” when



Fig. 2. Visual analysis task containing a balanced number of local details and global elements on each side of the image (vertical line added for descriptive purposes and was not presented to participants).

they saw a sad face and to say “sad” when they saw a happy face. Incorrect answers included the beginning of the wrong word even if it was not uttered fully.

Theory of mind. ToM skills in a social context were assessed using a French translation of the Child Faux Pas Test (Baron-Cohen et al., 1999; current version adapted from a translation by G. Roussarie, unpublished, available upon request). This test assesses higher-order ToM through understanding that the two characters in each situation might have different knowledge states, that the one saying the faux pas may affect the other emotionally, and that it might not have been said intentionally (Baron-Cohen et al., 1999). Understanding the faux pas also requires social reasoning based on social and moral norms in many cases.

Parent reports. Immediately prior to or during their children's session, parents were instructed to fill out a French translation of the 30-item Griffith Empathy Measure (Dadds et al., 2008), the Strengths and Difficulties Questionnaire (Goodman, 1997) to obtain a prosocial subscore, and the full Behavior Rating Inventory of Executive Function questionnaire (BRIEF; Gioia et al., 2002).

The empathy score and BRIEF inhibition score were used for the subsequent analysis. The prosocial subscore from the Strengths and Difficulties Questionnaire was used to measure correlations with the altruism scale and dictator game.

Control measure: Narrative skills. Narrative skills were assessed as a language-based control measure to confirm inclusion of participants in the final analysis. We used the Multilingual Assessment Instrument for Narratives (MAIN) stories test (Gagarina & Lindgren, 2020) in which children were randomly assigned to one of two comic-strip-style stories. Participants were asked to tell the story, and narrative details as well as character intentions were rated using the MAIN scorecard. Comprehension was assessed through 10 subsequent questions.

Altruism measures

Altruistic intention. Altruistic intention was measured using a translated, adapted, and validated version of the Altruism Scale for Children (ASC; Swank et al., 2020; Rose et al., 2023). The ASC measures hypothetical altruistic behaviors, referred to as altruistic intention in this study, through 30 short scenarios involving friends or classmates, family, neighbors, or animals. Each helping or sharing scenario includes three possible ways of reacting, with possible behaviors ranging from altruistic (3 points) to self-serving (1 point). Exploratory and confirmatory factor analyses were carried out with independent samples on the French adaptation and highlighted the existence of two distinct factors measuring altruistic intentions toward closer social circles ($\alpha = .82$) and toward more distant circles and animals ($\alpha = .79$), in a short-form 20-item version of the scale in French. This version was used in the current study.

Altruistic behavior. We adopted a dictator game paradigm to measure altruistic behavior used frequently and successfully with children (Benenson et al., 2007; Blake et al., 2015; Gummerum et al., 2010). Participants were told to choose 10 erasers from a box upon arrival to thank them for their participation. At one point during the session, they were told that another child of the same age was taking part in a separate study in which participants did not receive gifts. They were then offered the chance to share some erasers with this child (unnamed and without any visual cues as to the gender or appearance of the child) and were introduced to the dictator game hosted on an online platform created for this study.

In each of eight iterations, children were instructed to imagine that they had taken only the number of erasers noted on the far right column (see Fig. 3) and to imagine how many of these erasers they would be willing to give to the other child. They were then told that the computer would calculate the total of actual erasers to be given to the other child at the end of the session. The eight iterations were selected rather than a one-shot design to calculate an average donation ratio with both even and odd starting endowments and maximum donation amounts ranging from 2 to 9.

To simulate the implicit or explicit requests that are often observed in real life, children were given a slip of paper written by a child of their age asking to share their gifts (for a similar procedure, see

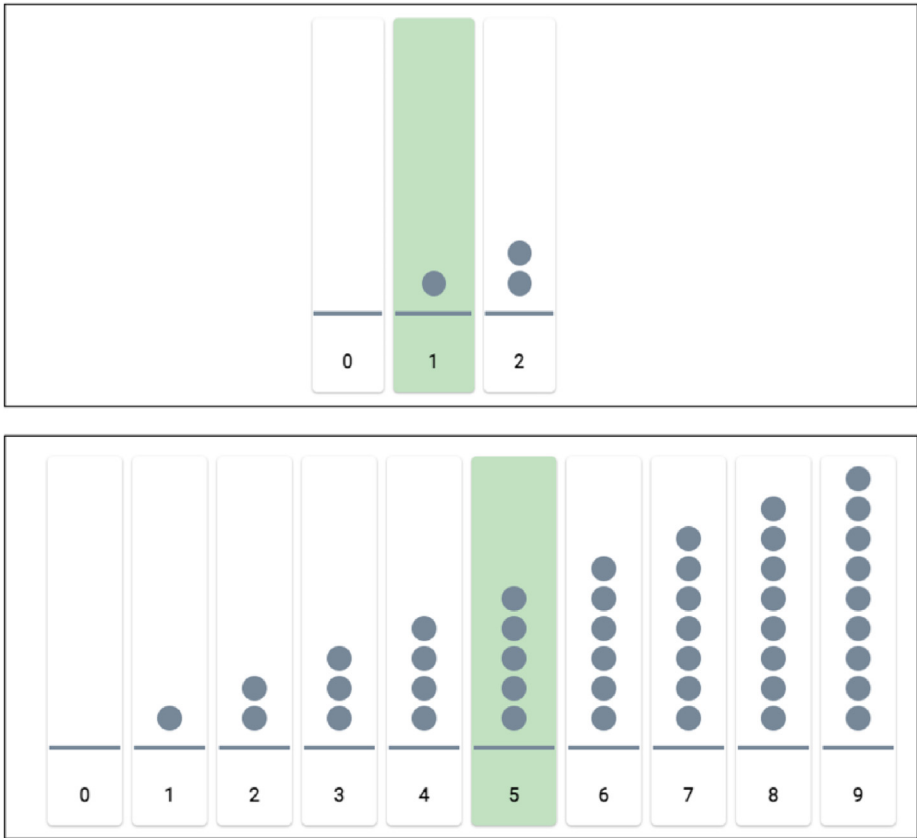


Fig. 3. Dictator game design presented in eight sequential iterations. Children were instructed to imagine that the total amount of erasers they had received was the number located on the far right of the screen and to click on the number depicting how many erasers they were willing to share with a child who had not received any. The experimenter left the room after controlling correct interpretation of instructions using an example.

Andreoni & Rao, 2011). The experimenter then left the room while participants completed the game. At the end of each session, the experimenter looked at a clock and told participants that the other child had already finished his or her session and that they could keep their 10 erasers this time. Participants were set to return for another round of testing as part of the larger altruism training program, so they were not debriefed at this point. After completing the full training program, participants were asked to rate how confident they were that the recipient was real, with 62% of children responding that they were somewhat to very confident in the existence of the recipient, 19% of children responding that they were unsure of the recipient's existence, and another 19% responding that they were somewhat to very confident that the recipient was hypothetical.

A summary of the variables reported in this study and a description of how they were measured are available in Table 2.

Data analysis

Data preprocessing

Language scores were used as a control measure to verify that participants could understand language-dense tasks such as the ToM task and altruism scale. All participants scored at least 14 of

Table 2

Reported variables and associated measures.

Variable	Measure	Output
Cognitive variables		
Visual analysis	Scene description task	Total visual analysis score
Selective attention	TEA-Ch attention battery, Sky Search subtest	Ratio of targets circled to total time on task in a scene with versus without distractors
Attentional flexibility	TEA-Ch attention battery, Opposite Worlds subtest	Difference between total time for tasks requiring attentional control and those with simple digit reading
Inhibition	Happy-Sad task	Total success score out of 20 attempts
Inhibitory difficulties	Parent-reported sub-questions on BRIEF	Raw total score
Theory of mind	Child Faux Pas Test (20 scenarios, 50% with and 50% without faux pas)	Ratio of correct test questions to correct control questions
Empathy	Parent-reported Griffith Empathy Measure	Raw total score
Narrative skills	MAIN, narrative elicitation task and comprehension questions	Total narrative + comprehension score (out of 20)
Altruistic variables		
Altruistic intention	Altruism Scale for Children, French version, shortened form (20 items)	Raw total score
Altruistic behavior	Dictator game with eight sequential iterations	Average ratio of erasers shared to total erasers across trials

Note. TEA-Ch, Test of Everyday Attention for Children; BRIEF, Behavior Rating Inventory of Executive Function questionnaire; MAIN, Multilingual Assessment Instrument for Narratives.

a possible 20 points on the language assessment and therefore were included in the subsequent analysis.

Selective attention, attentional flexibility, inhibition, parent-reported inhibition, ToM, and parent-reported empathy were entered as predictors, and the average donation ratio and altruism scale score were modeled as outcomes. All variables were standardized prior to inclusion in their respective regression models. Descriptive data on task scores per grade level are provided in [Table 3](#), and Pearson correlations between variables were computed and are displayed in [Table 4](#).

To prevent potential collinearity issues caused by the correlation between selective attention and attentional flexibility ($r = .43$), likely related to a developmental dependence (these two constructs develop concurrently in childhood; [Hanania & Smith, 2010](#)), we discarded the latter variable from the subsequent analysis. We decided not to group the selective attention and attentional flexibility scores into one composite attention score due to methodological proximity between our attentional flexibility and inhibition tasks and recruitment of shared as well as distinct brain regions in childhood ([Kerr-German & Buss, 2020](#)). For complete transparency related to this decision, the analysis script

Table 3

Mean standardized scores on each variable measured and standard deviations.

Variable measured	Score					
	Grade 3		Grade 4		Grade 5	
	Mean	SD	Mean	SD	Mean	SD
Dictator game	0.46	0.24	0.60	0.19	0.54	0.11
Altruism scale	0.80	0.16	0.86	0.10	0.84	0.10
Visual analysis	4.22	1.78	4.14	1.68	4.09	1.90
Selective attention	-4.81	1.31	-4.12	1.46	-3.14	1.38
Attentional flexibility	-8.89	5.14	-6.51	4.10	-6.04	2.87
Inhibitory difficulties	18.18	4.33	16.48	4.48	14.80	2.82
Inhibition	17.68	2.30	18.61	1.47	19.05	0.69
Theory of mind	0.67	0.35	0.82	0.28	0.73	0.25
Empathy	27.10	17.43	26.00	19.61	26.20	22.38

Table 4
Pearson correlations between variables.

	Age	Altruistic behavior	Altruistic intention	Visual analysis	Selective attention	Attentional flexibility	Inhibition	Inhibitory difficulties	Theory of mind	Empathy
Altruistic behavior	0.11	-								
Altruistic intention	0.01	0.35	-							
Visual analysis	0.19	0.15	-0.04	-						
Selective attention	0.22	0.11	0.24	-0.01	-					
Attentional flexibility	0.22	-0.03	-0.05	0.03	0.43*	-				
Inhibition	0.17	0.10	0.11	-0.14	0.38	0.26	-			
Inhibitory difficulties	-0.20	-0.29	-0.06	0.10	-0.22	-0.25	-0.29	-		
Theory of mind	0.11	-0.27	0.10	-0.05	0.08	0.22	<0.01	0.03	-	
Empathy	-0.05	0.06	-0.10	0.05	-0.13	0.04	-0.04	-0.28	0.06	

* $p < .05$.

including the dropped measure is available in the study's public repository (https://osf.io/cmvtg/?view_only=769793c0b37b41cebd47b941af06c13f).

Statistical analysis

A Bayesian approach was selected for data analysis for several reasons: to quantify the probability of observing an effect as well as to quantify the *uncertainty* related to our data and to provide an accurate and nuanced view of our results and, thus, a cautious interpretation and discussion (see van de Schoot et al., 2014, 2021, for comprehensive overviews of Bayesian statistics in research).

Bayesian frameworks are increasingly used in psychology research, including in evidence-based intervention and policy studies, for the statistical flexibility and robustness they can offer when properly implemented (Stamey et al., 2011; van de Schoot et al., 2017; Wagenmakers et al., 2018; Wagner & Gill, 2005). Our hypotheses, method, and Bayesian analysis plan were preregistered prior to launching the study and are available in the study's public repository.

Data analysis was carried out using R (R Core Team, 2022), the *easystats* environment (Makowski et al., 2019) and the “psych” (Revelle, 2016) and “brms” (Bürkner, 2017) packages.

Following the pre-registration plan, we assessed the contribution of visual analysis, attention, inhibition, ToM, and empathy on altruistic intention and behavior. Regression models were fitted under the Bayesian framework, with mildly informative *t* Student priors (3 degrees of freedom to accommodate potential outliers; Vanhatalo et al., 2009). Additional information, including distribution plots and model fit measures, is available along with the full R analysis script in the study's public repository.

Age and grade were also added as predictors in the model; grade was included given that children in one grade level may vary in age but are recipients of the same school curriculum and take part in grade-specific peer interactions. We computed the 95% credible interval (CI) using highest density intervals, which contain 95% of the posterior distribution. Following the approach used in Tunçgenç et al. (2021), when the effect was not found to be significantly different from 0 (i.e., when 0 was within the 95% CI), we also conducted a Bayesian hypothesis test (using the “brms” package) to quantify the probability that the effect was strictly > 0 with Bayes factors (BF_{10}) to provide more nuance to “non-significant.” For the sake of comparison, Bayesian hypothesis tests were also conducted when the 95% CI did not include 0. Following our hypothesis, when we test the positive contribution of a cognitive ability to the model, we obtain an evidence ratio (BF_{10}). We consider that an evidence ratio > 10 provides strong evidence for our hypothesis, meaning that an effect of the cognitive ability was at least 10 times more likely than a null effect.

Results

Altruistic intention

A collinearity check found a low risk of multicollinearity between variables, notably between age and grade level for all altruistic intention models tested. The regression model predicting altruistic intention (Table 5) suggested that a difference in grades, and more specifically belonging to the fourth grade compared with the third grade, had the strongest impact on altruistic intention ($\beta = 0.45$, 95% CI $[-0.23, 1.16]$), followed by a positive impact of selective attention ($\beta = 0.20$, 95% CI $[-0.10, 0.50]$).

To get a clearer picture of how grade level affects altruistic intention, we then tested another model, referred to as the altruistic intention interaction model (Table 5), including interactions between grade level and selective attention as well as between grade level and visual analysis, a separate but related process given the visual nature of our selective attention task (Fig. 4).

This model confirmed the importance of selective attention in altruistic intention, with a contribution modulated by grade level. The positive effect of selective attention scores in the lowest grade ($\beta = 0.67$, 95% CI $[0.19, 1.14]$) was significantly attenuated in the fourth grade ($\beta = -0.84$, 95% CI $[-1.46, -0.21]$), but not significantly in the fifth grade ($\beta = -0.41$, 95% CI $[-1.07, 0.26]$). After selective attention, the only cognitive skill to predict altruistic intention in this model was visual analysis, also with variations across grade levels. Results highlight a positive effect on altruistic intention of the

Table 5
Of the Bayesian linear regression predicting altruistic intention scores.

Predictor	β	95% Credible interval	Posterior inclusion probability	BF ₁₀
Altruistic intention model: $R^2 = 16.9\%$				
Visual analysis	0.01	[−0.26, 0.28]	53.36%	1.14
Selective attention	0.20	[−0.10, 0.50]	90.92%	10.00
Inhibition	−0.0005	[−0.29, 0.29]	51.40%	1.06
Inhibitory difficulties	−0.02	[−0.31, 0.26]	56.47%	1.30
Theory of mind	0.02	[−0.25, 0.29]	57.12%	1.33
Empathy	−0.05	[−0.33, 0.23]	63.67%	1.75
Age	−0.14	[−0.51, 0.23]	77.63%	3.47
Grade Level A (third)	−0.23	[−0.78, 0.31]	80.20%	4.05
Grade Level B (fourth vs. third)	0.45	[−0.23, 1.16]	89.99%	8.99
Grade Level C (fifth vs. third)	0.26	[−0.62, 1.20]	71.45%	2.50
Altruistic intention interaction model: $R^2 = 29.7\%$				
Visual Analysis * Grade Level A	0.26	[−0.15, 0.66]	89.68%	8.69
Visual Analysis * Grade Level B	−0.39	[−0.99, 0.21]	90.57%	9.60
Visual Analysis * Grade Level C	−0.27	[−0.86, 0.31]	82.08%	4.58
Selective Attention * Grade Level A	0.67	[0.19, 1.14]	99.69%	316.00
Selective Attention * Grade Level B	−0.84	[−1.46, −0.21]	99.50%	197.00
Selective Attention * Grade Level C	−0.41	[−1.07, 0.26]	88.43%	7.64
Inhibition	−0.03	[−0.30, 0.25]	57.46%	1.35
Inhibitory difficulties	−0.03	[−0.32, 0.25]	59.74%	1.48
Theory of mind	0.01	[−0.25, 0.27]	53.22%	1.14
Empathy	−0.07	[−0.33, 0.19]	69.89%	2.32
Age	−0.05	[−0.41, 0.32]	60.62%	1.54
Grade Level A	0.04	[−0.50, 0.59]	55.20%	1.23
Grade Level B	0.17	[−0.51, 0.85]	69.30%	2.26
Grade Level C	−0.10	[−1.06, 0.84]	58.45%	1.41

interaction between visual analysis scores and the lowest grade level ($\beta = 0.26$, 95% CI [−0.15, 0.66]) and a negative interaction with the two higher grade levels (for the difference between the fourth and third grades, $\beta = -0.39$, [−0.99, 0.21]; for the difference between the fifth and third grades, $\beta = -0.27$, 95% CI [−0.86, 0.31]).

Other cognitive capabilities did not contribute in a significant manner to the model, and neither did empathy ($\beta = -0.07$, 95% CI [−0.33, 0.19], with a low probability of direction of 69.89%).

Subsequent hypothesis testing, however, showed only moderate evidence of an effect of the interaction between visual analysis and grade level in the first two grades (a positive effect, BF₁₀ = 8.69, and a negative effect, BF₁₀ = 9.60, respectively) and showed relatively low-moderate evidence of a negative effect of the interaction between visual analysis and fifth grade, the highest grade in the study (BF₁₀ = 4.58). A hypothesis test of the negative effect of the interaction between selective attention and the highest grade yielded moderate evidence (BF₁₀ = 7.64).

A model including the dropped variable, attentional flexibility, was also run to ensure full transparency in our analysis. Hypothesis tests yielded similar evidence: strong evidence of a positive effect of the interaction between grade level and visual analysis skills in the youngest grade and strong evidence of a negative effect of this interaction for the next two grade levels (BF₁₀ = 27.65, BF₁₀ = 18.69, and BF₁₀ = 10.04, respectively) and strong evidence of a positive effect of the interaction between grade level and selective attention skills in the youngest grade and strong evidence of a negative effect of this interaction for the next two grade levels (BF₁₀ = 2856.14, BF₁₀ = 407.16, and BF₁₀ = 21.25, respectively). The only significant difference with the main model was strong evidence found of a negative effect of the interaction between grade level and attentional flexibility in the youngest grade only (BF₁₀ = 44.66).

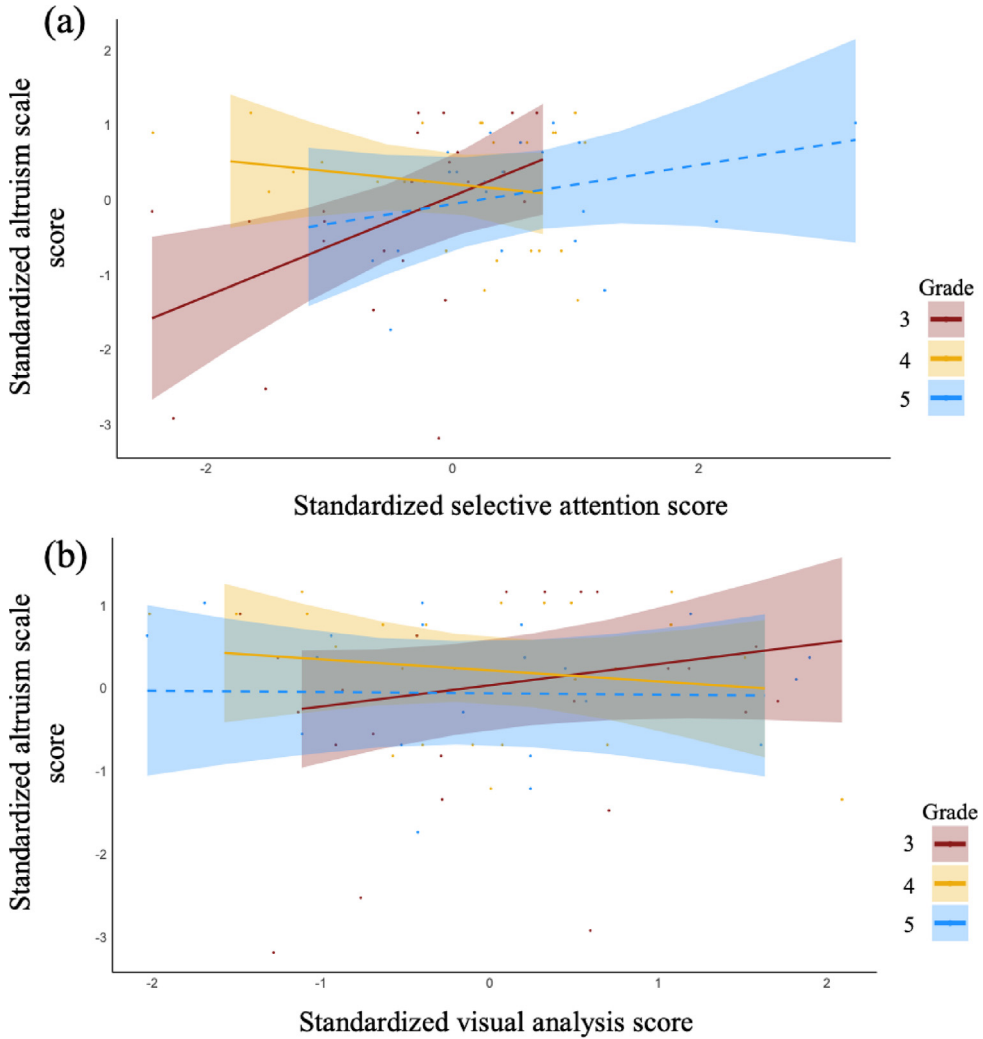


Fig. 4. (A) Bayesian regression model estimate of the interaction between selective attention and grade and its association with average score on the altruism scale. (B) Bayesian regression model estimate of the interaction between visual analysis and grade and its association with average score on the altruism scale. Solid lines represent interactions for which there is strong evidence ($BF_{10} > 10$), and dashed lines represent interactions for which there is moderate evidence ($4 < BF_{10} < 10$).

Exploratory analysis: Cognitive predictors for each altruistic intention factor

We conducted an additional exploratory analysis to determine whether distinct cognitive capabilities predicted scores on the two different factors of the altruistic intention scale. Factor 1, which measured altruistic intention toward the close social circle, and Factor 2, which measured altruistic intention toward more distant circles and animals, were correlated ($r = .70$, $p < .001$).

In line with results on the full altruistic intention scale, we found that selective attention had a significant positive effect on altruistic intention in the youngest grade toward close circles ($\beta = 0.63$, 95% CI [0.11, 1.41]) and more distant social circles ($\beta = 0.64$, 95% CI [0.14, 1.14]). In the following grade, this

effect was significantly dampened toward both close social circles ($\beta = -0.84$, 95% CI $[-1.56, 0.12]$) and those more distant ($\beta = -0.83$, 95% CI $[-1.54, -0.13]$).

We also found that in the highest grade stronger scores in the ToM task significantly negatively predicted altruistic intention, but only toward distant circles and animals ($\beta = -0.54$, 95% CI $[-1.21, 0.14]$).

Altruistic behavior

A collinearity check found a low risk of multicollinearity between variables, notably between age and grade level for all altruistic behavior models tested. Results of the model predicting altruistic behavior (altruistic behavior model; Table 6), indicated a strong positive contribution of grade level to altruistic behavior, notably for the fourth grade compared with the third grade ($\beta = 0.83$, 95% CI $[0.18, 1.47]$). Scores on the Faux Pas Test, which assessed a facet of ToM, were the strongest cognitive predictor of altruistic behavior in the model, displaying a negative effect ($\beta = -0.31$, 95% CI $[-0.55, -0.06]$).

Disinhibition was also found to be a negative predictor of altruistic behavior, but only in terms of behavioral disinhibition reported by parents ($\beta = -0.26$, 95% CI $[-0.52, 0.00]$), with no effect of cognitive inhibition tested in children in the laboratory ($\beta = -0.01$, 95% CI $[-0.27, 0.24]$).

To determine whether differences in grade differentially altered the effect of ToM (measured by the Faux Pas Test) and inhibition on altruistic behavior, we implemented another regression model, the altruistic behavior interaction model (Table 6), including faux pas scores, parent-reported disinhibition, and their interactions with grade level as predictor variables in addition to the other cognitive and social-emotional variables.

We found that with regard to ToM, only the interaction of faux pas scores and the lowest grade influenced altruistic behavior and negatively so ($\beta = -0.42$, 95% CI $[-0.75, -0.08]$). The interaction

Table 6
Results of the Bayesian linear regression predicting altruistic behavior scores.

Predictor	β	95% Credible interval	Posterior inclusion probability	BF ₁₀
Altruistic behavior model: $R^2 = 31.4\%$				
Visual analysis	0.18	$[-0.07, 0.42]$	92.61%	12.50
Selective attention	0.05	$[-0.22, 0.31]$	64.48%	1.81
Inhibition	-0.01	$[-0.27, 0.24]$	54.64%	1.20
Inhibitory difficulties	-0.26	$[-0.52, 0.00]$	97.58%	40.20
Theory of mind	-0.31	$[-0.55, -0.06]$	99.31%	145.00
Empathy	-0.0006	$[-0.24, 0.26]$	52.20%	1.09
Age	-0.09	$[-0.42, 0.25]$	69.80%	2.31
Grade Level A (third)	-0.40	$[-0.92, 0.11]$	93.88%	15.30
Grade Level B (fourth vs. third)	0.83	$[0.18, 1.47]$	99.30%	143.00
Grade Level C (fifth vs. third)	0.36	$[-0.49, 1.22]$	79.42%	3.86
Altruistic behavior interaction model: $R^2 = 39.5\%$				
Visual Analysis * Grade Level A	0.22	$[-0.16, 0.61]$	86.99%	6.69
Visual Analysis * Grade Level B	-0.21	$[-0.77, 0.35]$	77.36%	3.42
Visual Analysis * Grade Level C	-0.02	$[-0.61, 0.55]$	52.72%	1.11
Inhibitory Difficulties * Grade Level A	-0.50	$[-0.87, -0.13]$	99.56%	229.00
Inhibitory Difficulties * Grade Level B	0.36	$[-0.14, 0.86]$	92.02%	11.50
Inhibitory Difficulties * Grade Level C	0.76	$[0.01, 1.50]$	97.71%	42.70
Theory of Mind * Grade Level A	-0.42	$[-0.75, -0.08]$	99.12%	113.00
Theory of Mind * Grade Level B	0.08	$[-0.44, 0.60]$	61.55%	1.60
Theory of Mind * Grade Level C	0.22	$[-0.39, 0.81]$	76.21%	3.20
Selective attention	0.05	$[-0.24, 0.33]$	63.81%	1.76
Inhibition	-0.02	$[-0.28, 0.24]$	54.75%	1.21
Empathy	0.08	$[-0.18, 0.34]$	72.58%	2.65
Age	-0.0009	$[-0.36, 0.33]$	52.29%	0.91
Grade Level A	-0.26	$[-0.80, 0.26]$	83.89%	5.21
Grade Level B	0.70	$[0.05, 1.36]$	98.20%	54.70
Grade Level C	0.37	$[-0.52, 1.26]$	79.84%	3.96

between parent-reported disinhibition had a negative effect on altruistic behavior in the lowest grade ($\beta = -0.50$, 95% CI $[-0.87, -0.13]$), meaning that stronger behavioral inhibition skills were related to higher donations in the dictator game. This trend, however, became inverted and results highlight a positive effect of the interaction between disinhibition and grade level for higher grades compared with the lower grade ($\beta = 0.36$, 95% CI $[-0.14, 0.86]$ for the fourth grade and $\beta = 0.76$, 95% CI $[0.01, 1.50]$ for the fifth grade). These effects are illustrated in Fig. 5. A grade-related positive trend in altru-

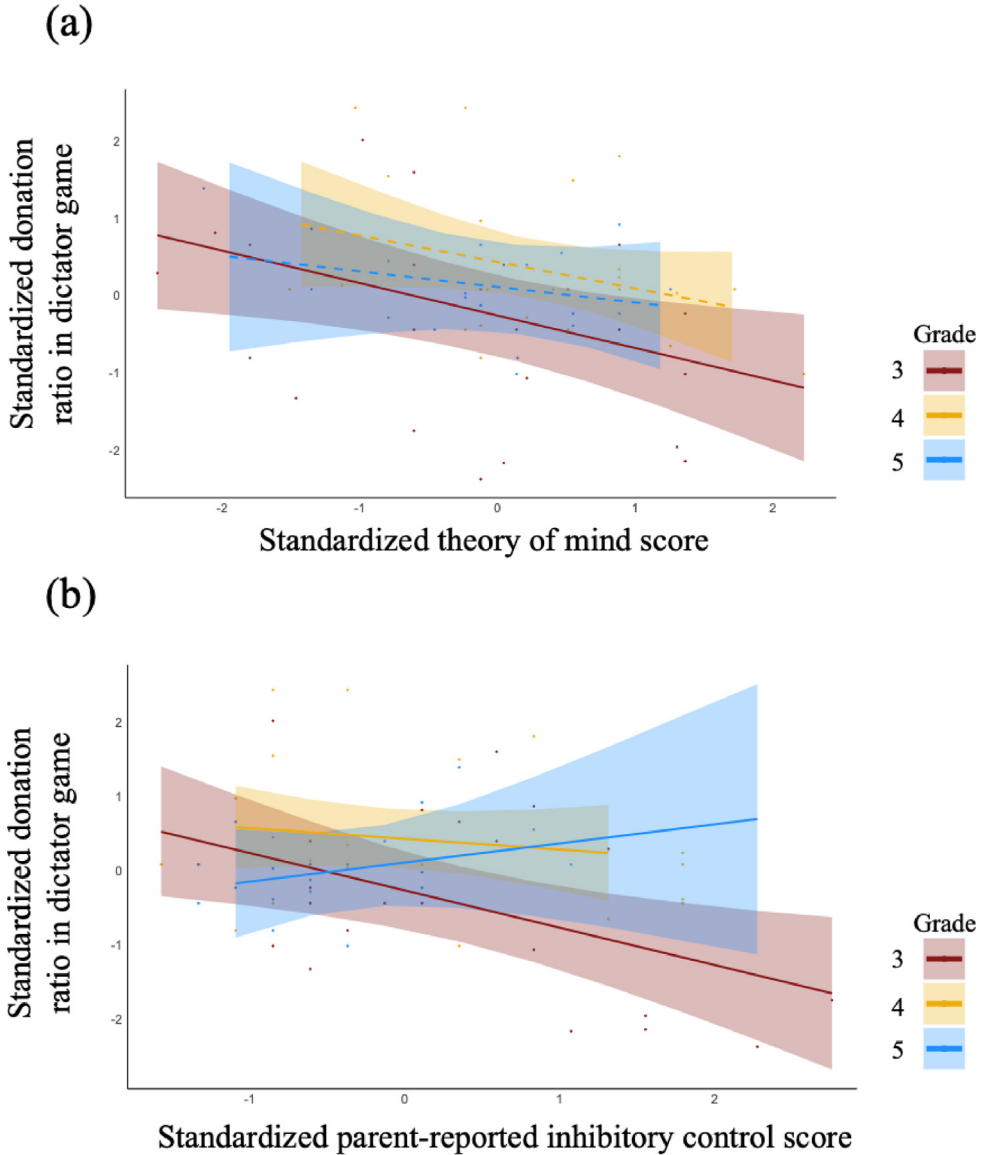


Fig. 5. (A) Bayesian regression model estimate of the interaction between ToM measured by a faux pas task and grade and its association with average donation in a dictator game. (B) Bayesian regression model estimate of the interaction between parent-reported inhibitory control and grade and its association with average donation in a dictator game. Solid lines represent interactions for which there is strong evidence ($BF_{10} > 10$), and dashed lines represent interactions for which there is little to no evidence ($BF_{10} < 4$).

istic behavior remained when controlling for cognitive variables, notably with a positive effect of the middle grade versus the lowest grade ($\beta = 0.70$, 95% CI [0.05, 1.36]).

The remaining cognitive capabilities included in the model did not show an effect on altruistic behavior, as illustrated in Table 6, and we did not find an effect of empathy on altruistic behavior ($\beta = 0.08$, 95% CI [-0.18, 0.34], with a probability of direction of 72.58%). Further examination of Bayes factors for predictors with confidence intervals including zero confirmed that evidence did not support an effect of ToM on altruistic behavior when interacting with the two higher grades ($BF_{10} = 1.60$ and $BF_{10} = 3.20$, respectively). However, hypothesis testing provided strong evidence supporting a positive effect of disinhibition on altruistic behavior when interacting with the two higher grades ($BF_{10} = 11.50$ and $BF_{10} = 42.70$, respectively). Finally, we found moderate evidence of a positive effect of visual analysis skills on altruistic behavior, but only when interacting with the lowest grade ($BF_{10} = 6.69$), with no evidence of an effect in higher grades.

Again, a model including attentional flexibility was run to ensure full transparency in our analysis. Hypothesis tests yielded similar results: strong evidence of a negative effect of the interaction between grade level and parent-reported inhibition difficulties in the youngest grade and strong evidence of a positive effect of this interaction for the next two grade levels ($BF_{10} = 297.51$, $BF_{10} = 9.76$, and $BF_{10} = 36.74$, respectively) and strong evidence of a negative effect of the interaction between grade level and ToM in the youngest grade ($BF_{10} = 54.71$).

Discussion

The current study aimed to gain a better understanding of the comparative effects of various cognitive capabilities and empathy on altruistic intention and behavior. Results only partially validated our initial hypothesis yet provide intriguing new data on what could be referred to as a cognition-altruism hypothesis.

We found evidence that distinct cognitive abilities differentially drove altruistic intention and behavior. Selective attention and, to a lesser extent, visual analysis positively predicted altruistic intention in the lowest (third) grade; understanding of intentionality in social settings, a process involved in ToM, was found to be a negative predictor, and parent-reported behavioral inhibition was found to be a positive predictor, of altruistic sharing in the youngest grade. Interestingly, we also found that the effects of these cognitive abilities changed direction as grade levels increased. Compared with the youngest grade, increases in selective attention were associated with lower altruistic intention scores in higher grades, and behavioral inhibition drove altruistic giving toward a baseline average value as capacities increased. This suggests that although certain cognitive capabilities such as selective attention, visual analysis, and behavioral inhibition support increased altruistic intention and behavior in the third grade, they tend to limit it in later grades. Surprisingly, empathy was not found to predict altruistic intention or behavior throughout the study. Specific salient results are discussed in the following paragraphs.

Cognitive drivers of altruistic intention

The effects of selective attention and visual analysis on altruistic intention in the current study imply questioning what our altruism scale measures. The scale consists of hypothetical daily life situations in which participants indicate how they would act from three possible choices. Increased visual analysis and selective attention in daily life may lead to more frequent encounters and familiarity with situations in which helping or sharing are warranted given that they are paid attention to, and analyzed correctly, more often. This, however, held only for the youngest grade level. In the two higher grades, selective attention and, to a lesser extent, visual analysis negatively predicted altruistic intention. This may have a methodological explanation; the visual analysis task and the TEA-Ch selective attention task might not be sensitive enough in higher grades in typically developing children, although no ceiling effect of these two tasks was reported upon detailed investigation. Another possibility, drawing from Hao's (2017) work on cool and hot inhibition, is that more socially centered attentional processes take precedence in later grades. This hypothesis could be supported by

the ongoing development of attention networks, including orienting and alerting capabilities, well into young adulthood (Boen et al., 2021), and the fact that visual attention is oriented based on social goals in interactions, so increased visual analysis and attentional abilities may indeed support stronger social reasoning (Teoh et al., 2020).

Cognitive drivers of altruistic behavior

Our results regarding cognitive drivers of altruistic behavior, notably the impact of inhibition and processes involved in ToM on subsequent altruistic sharing, followed a similar pattern.

Our findings regarding inhibition should be interpreted on several different levels. First, we found no effect of cognitive inhibition as measured by an experimental task, the Happy–Sad task, on altruistic giving. The Happy–Sad task was selected for its sensitivity even with adolescents and adults (Lagattuta et al., 2011), but score distribution was somewhat skewed and would have benefitted from the use of a more difficult task for this age group. Parent-reported disinhibition scores, however, offer interesting insight. Echoing the findings linking selective attention to altruistic intention, we found that parent-reported inhibition abilities had a positive impact on altruistic sharing in the younger grade and had a limiting effect in both higher grades. Inhibition is a key process within more global executive functioning that underlies decision making and plays a role in overcoming cognitive errors throughout the lifespan (Borst et al., 2015; Sakagami et al., 2006). Our findings are in line with the notion that inhibition skills support prosocial and altruistic behavior in younger children by helping them to exert control over predispositions toward selfish behavior to subsequently activate social sharing norms (Aguilar-Pardo et al., 2013). As for the inversion of this effect in subsequent grade levels, acting prosocially may depend on a necessary level of inhibition, but increased inhibition may support more socially strategic behavior and therefore more limited giving to a hypothetical and unknown other, especially in a private setting (Buhrmester et al., 1992).

We found that in the youngest grade increases in ToM were related to more limited altruistic sharing. Cowell and colleagues (2015) came to a similar conclusion as our study, where passing a false-belief task was related to less altruistic sharing in a sample of preschoolers. Our results, as well as a body of empirical work, suggest that the processes underlying ToM development help children to reason more strategically about how, with whom, and to which extent to enact altruistic behavior: Children behave altruistically more selectively in middle childhood, preferring to share with friends and helping based on the level of information they share with a recipient (Paulus & Moore, 2014; Siposova et al., 2021; Weller & Hansen Lagattuta, 2013). The negative relationship between the facet of ToM we measured here and altruistic sharing, however, stands out only for the youngest grade. Recent models confirm the existence of distinct types of advanced ToM, namely social reasoning, reasoning about ambiguity, and understanding social norm transgressions, that increase throughout middle childhood (Osterhaus & Koerber, 2021; Osterhaus et al., 2016). Adding ToM tasks that measure the remaining two factors may shed greater light on the relationship with altruistic behavior in later grades.

Several participants in our study explicitly asked how much others decided to give in the dictator game, hinting that they may have engaged in social comparison with hypothetical others when deciding how many erasers to share. Reputational concerns play an explicit role in prosocial decision making from 8 years of age, which was roughly the age of our youngest grade in this study (Engelmann & Rapp, 2018). Attention to reputation is driven by increases in ToM capabilities and affects altruistic behavior by limiting actions that would be considered overly generous compared with peers (Banerjee, 2002; Tasimi et al., 2015). Social comparison, referred to in this case as “do-gooder derogation,” leads to negative appreciations of overt generosity (Tasimi et al., 2015). Given that perspective taking can lead to metaperceptual inferences, where the actors focus on what the recipients may think about them and their actions (going beyond simple mental state inference), children with stronger abilities underlying ToM may engage in this type of complex inference more easily, leading to stronger social comparison and subsequently to limited altruistic behavior (Sassenrath et al., 2016). Altruism toward members of our social group involves more reputational mechanisms than altruism toward nonmembers (Mifune et al., 2010), so increased social–cognitive reasoning capabilities may lead chil-

dren to prioritize altruistic intention and behavior depending on who the recipient is and what they stand to gain from their actions.

As such, it appears that altruism itself changes throughout middle childhood, perhaps more spontaneously and based on perceptual information in younger grades and evolving into a more complex decision that involves weighing various factors in later childhood, which would help to explain why certain cognitive skills supported, and then curbed, altruistic intention and behavior in our study. In addition to using cognitive tasks that better capture the more complex nature of cognitive processes in later grades, going even further in adapting measures of altruism to this rapidly changing social landscape may prove to be fruitful.

Empathy, an outlier

In contrast to empirical findings supporting the empathy–altruism hypothesis, we did not find support for a role of empathy in predicting altruistic intention or behavior with the scales and paradigms used in this study (Batson et al., 1988; Cialdini et al., 1987; Eisenberg et al., 1991; Estrada, 1995; Hoffman, 2000; Malti et al., 2009; Nagar et al., 2020). Our findings are in line with certain studies in which empathy does not predict altruistic giving or, on the contrary, decreases it (Kozloff et al., 2021; Schaefer et al., 2021). The use of a parent-report measure may point to a bias in responses; however, this was not the case given that our results on this measure were normally distributed, and parent-report measures have been used to underline positive contributions of empathy to prosocial behavior in the past (see Nagar et al., 2020).

Outstanding challenges

Our study did not investigate the full scope of factors included in the SIP model, including episodic memory, an important factor in the model that has been linked to prosocial decision making and willingness to help in adults (Crick & Dodge, 1994; Gaesser & Fowler, 2020; Gaesser & Schacter, 2014). Future research should also include this aspect for a fuller consideration of the model's application to altruistic behavior. In addition, we looked at only one facet of actual altruistic behavior, donation. Measuring real-life helping and comforting will be necessary in subsequent studies to gain a deeper understanding of how cognitive processes differentially support altruistic behavior. Our measure of altruistic sharing was also somewhat hypothetical given that we did not ask participants to physically give their erasers away and indicated at the end of the session that the recipient had already finished the study and left the building. Our aim was to create a task that went beyond a one-shot measure and in doing so led to a compromise on the reality of our design. Although the dictator game has been widely used in past developmental research (Benenson et al., 2007; Gummerum et al., 2010; Harbaugh & Krause, 2000), it fails to test assumptions about what happens in real-life interactions. Testing these “ecological commitments” (Dahl, 2017) and how cognition and altruism relate in daily situations will be necessary in future research.

Concluding remarks

Our results on the impact of visual analysis, attention, inhibition, and ToM support the notion that cognitive capabilities play a crucial role in social decision making and behavior. Visual analysis and attention abilities appear to scaffold intention to act altruistically in younger children, contributing to limited—and quite possibly more strategic—altruistic consideration in later childhood. Behavioral inhibition abilities follow a similar pattern, supporting altruistic sharing at first and contributing to more nuanced limited sharing in older children. Increased ability to infer intentionality, a process involved in ToM, was found to limit altruistic sharing in younger children. By highlighting that different cognitive capabilities underlie specific types of altruistic intention and behavior, and affect them differently throughout middle childhood, our findings add to growing calls to consider cognitive and social–emotional development concurrently in practical applications (Duraiappah et al., 2022). We also aimed to underline, through this research, that social–emotional learning programs are not

one-size-fits-all; they need to be carefully adapted according to grade level, going beyond traditional elementary school versus middle and high school distinctions. By adapting program content to developmental specificities from one grade to another, they can target cognitive processes crucial to prosocial behavior more accurately. It is also important that they consider growing reputational concerns throughout childhood, general peer contexts, and the complex social and strategic reasoning that cognitive developments allow. Cognition is crucial, not to lighten the burden of another indiscriminately but rather for children to learn to use altruistic behavior efficiently—and usefully—throughout the lifespan.

Data availability

Data is available in a public repository at the following address: https://osf.io/cmvtg/?view_only=769793c0b37b41cebd47b941af06c13f.

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